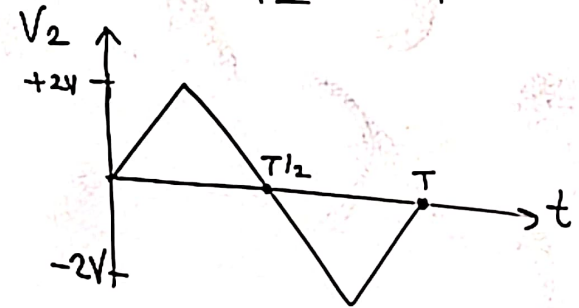
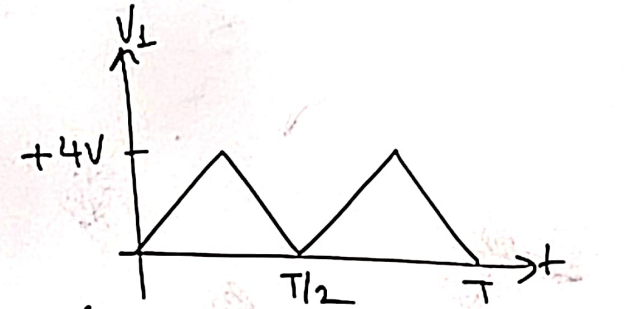
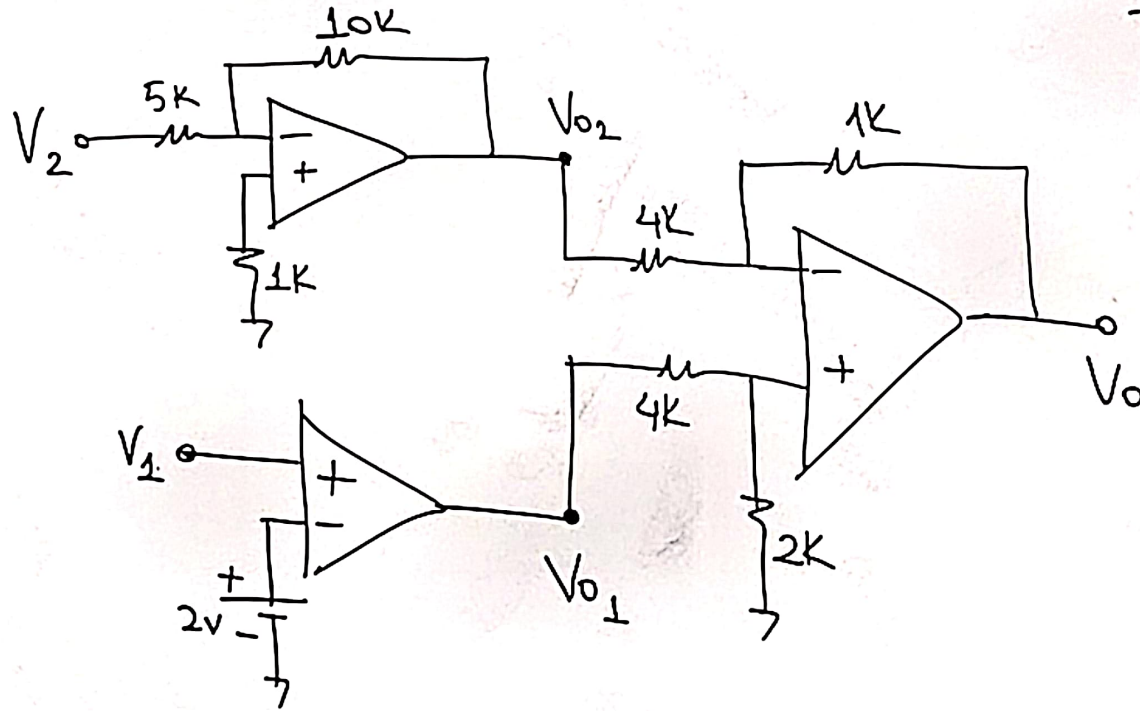


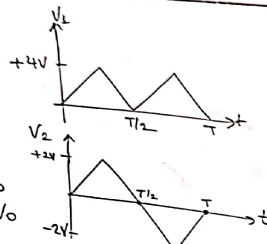
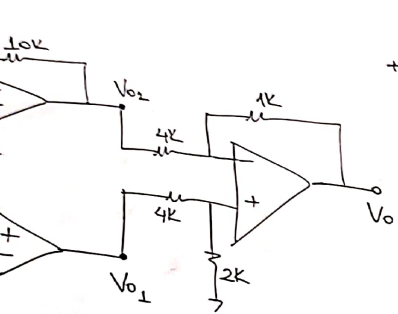
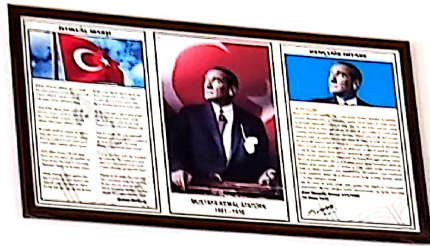
2024 - Büt.



$$\pm V_{cc} = \pm 12V$$

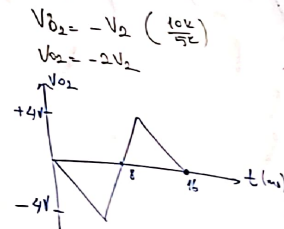
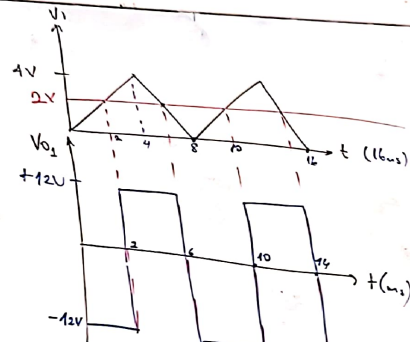
$$T = 16ms$$

V_{01}, V_{02}, V_0 'ı örnekli olarak çiz.

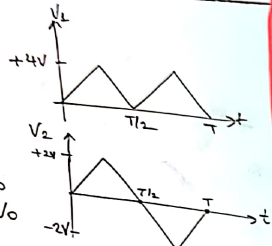
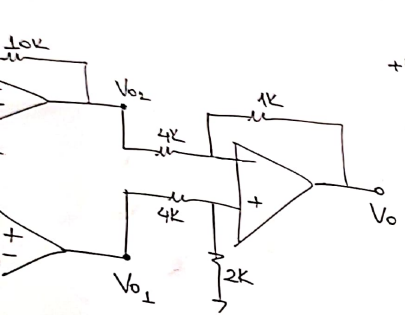


$$\pm V_{CC} = \pm 12V$$
$$T = 16ms$$

V_{01}, V_{02}, V_{01}' örneği olarak girer.

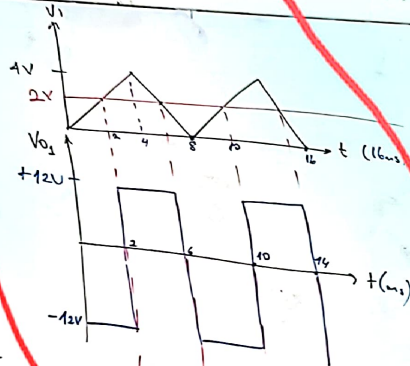


$$V_{02} = -V_2 \left(\frac{10k}{4k} \right)$$
$$V_{02} = -2V_2$$

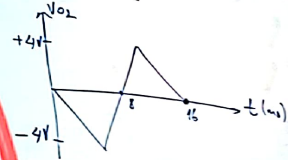


$$\pm V_{cc} = \pm 12V$$
$$T = 16ms$$

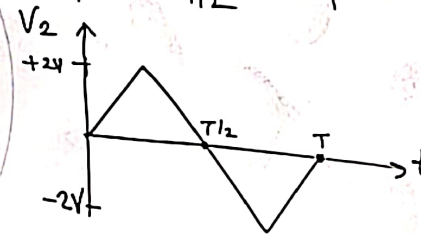
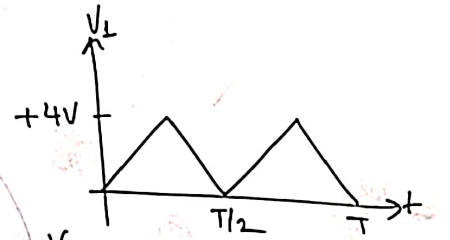
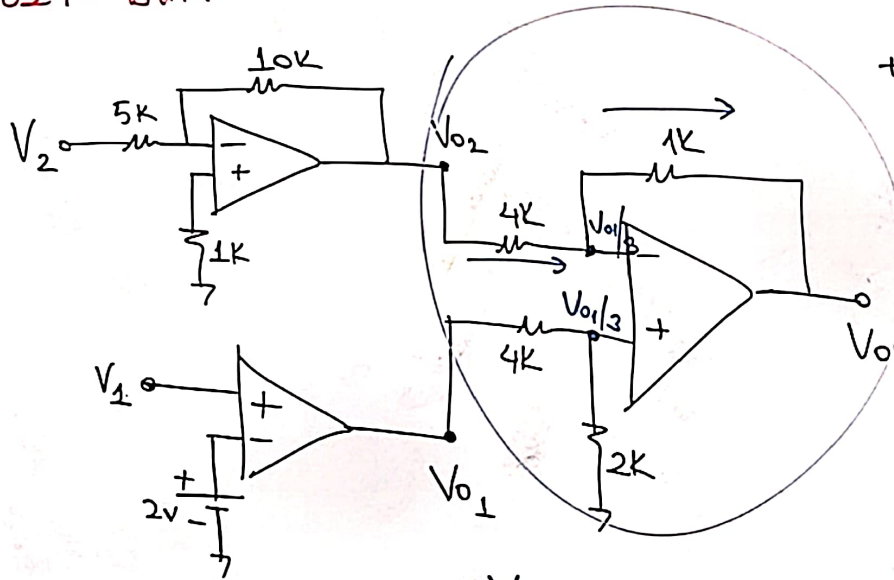
V_{01}, V_{02}, V_{01}' özetli olarak çiz.



$$V_{02} = -V_2 \left(\frac{10k}{4k} \right)$$
$$V_{02} = -2V_2$$



2024 - Büt.



$$\pm V_{cc} = \pm 12V$$

$$T = 16ms$$

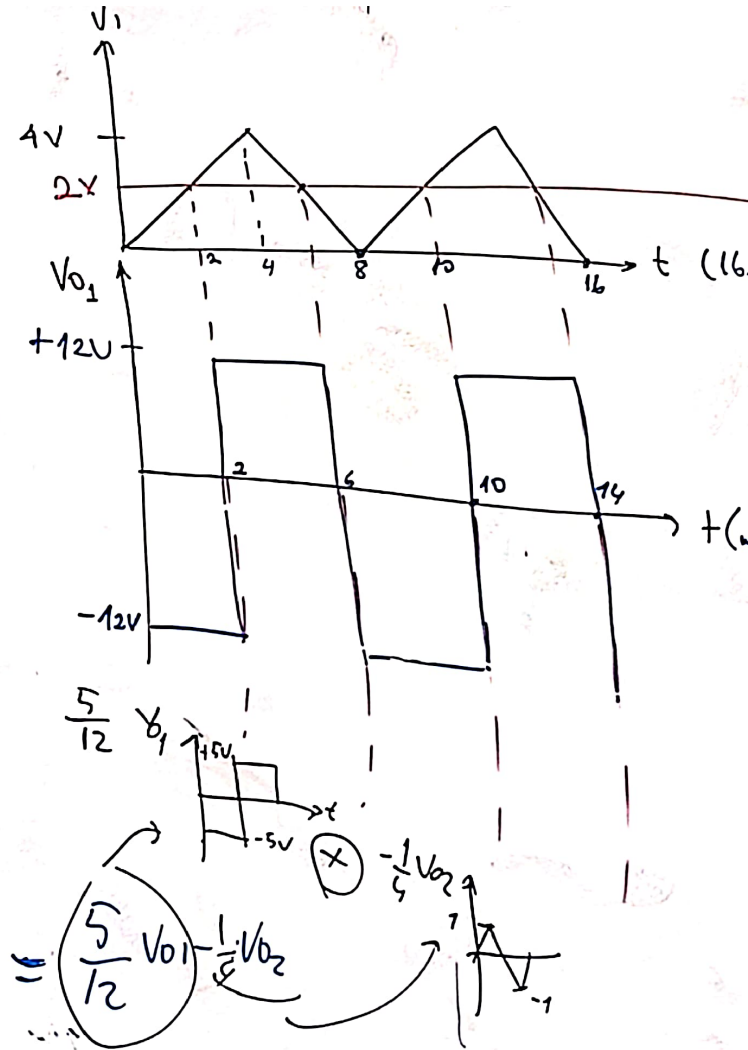
V_{01}, V_{02}, V_0 0 olarak girer.

$$\frac{V_{01}}{6K} \cdot 2K = V_+$$

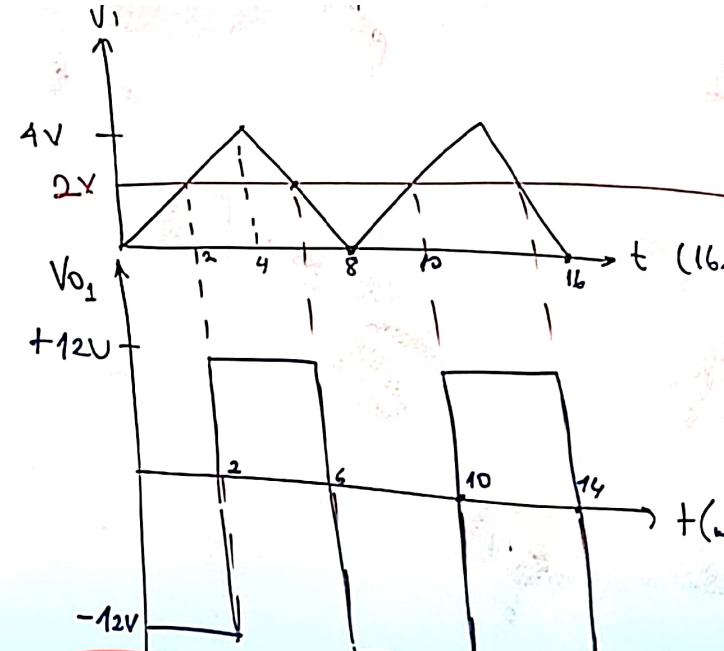
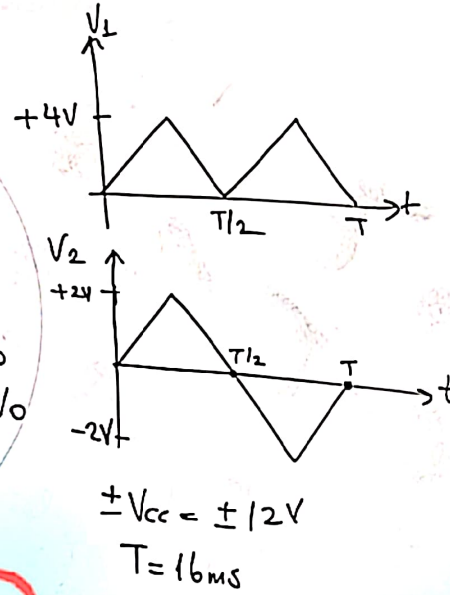
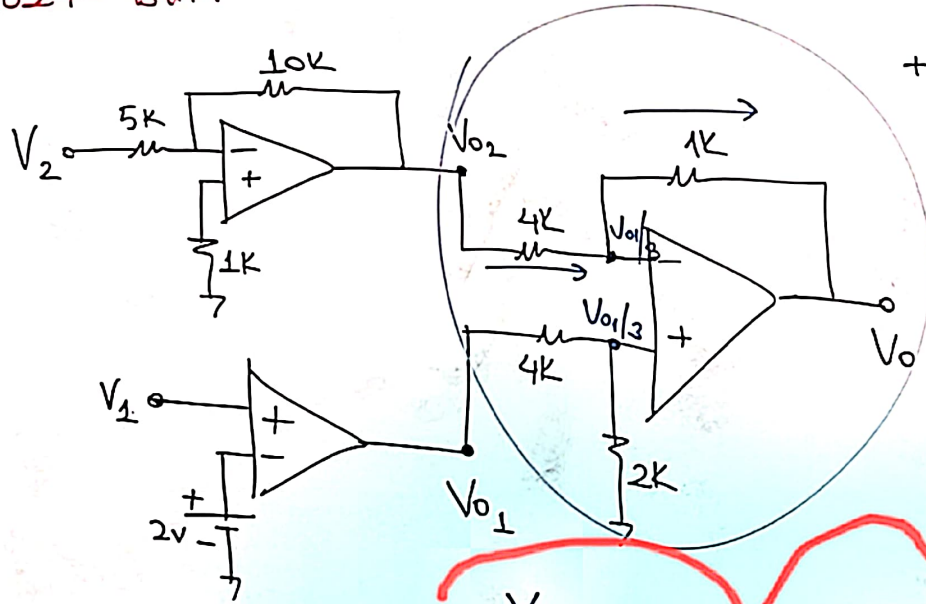
$$\frac{V_{02} - V_{01/3}}{4K} = \frac{V_{01/3} - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$



2024-Büt.

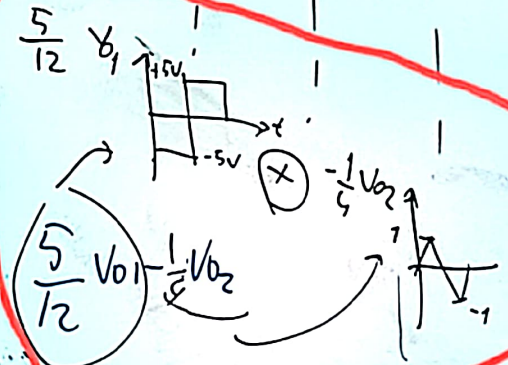


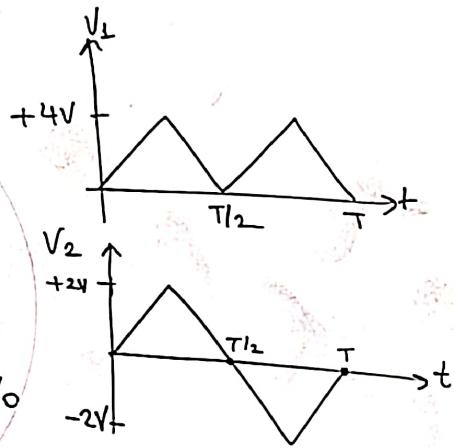
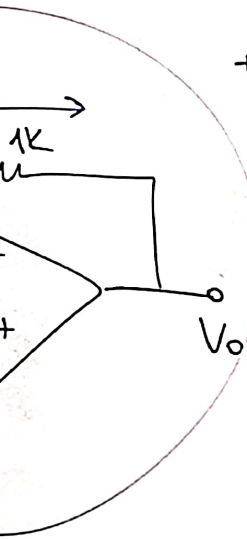
$$\frac{V_{01}}{6K} \cdot 2K = V_+$$

$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

$$4V_0 = \frac{5}{3}V_{01} - V_{02} \Rightarrow V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right)$$





$$\pm V_{CC} = \pm 12V$$

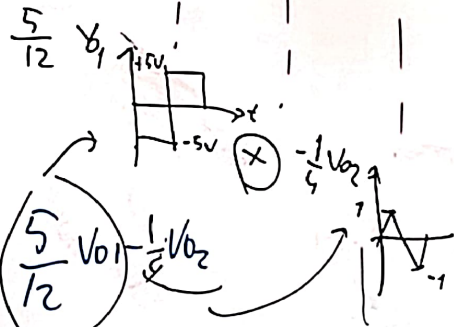
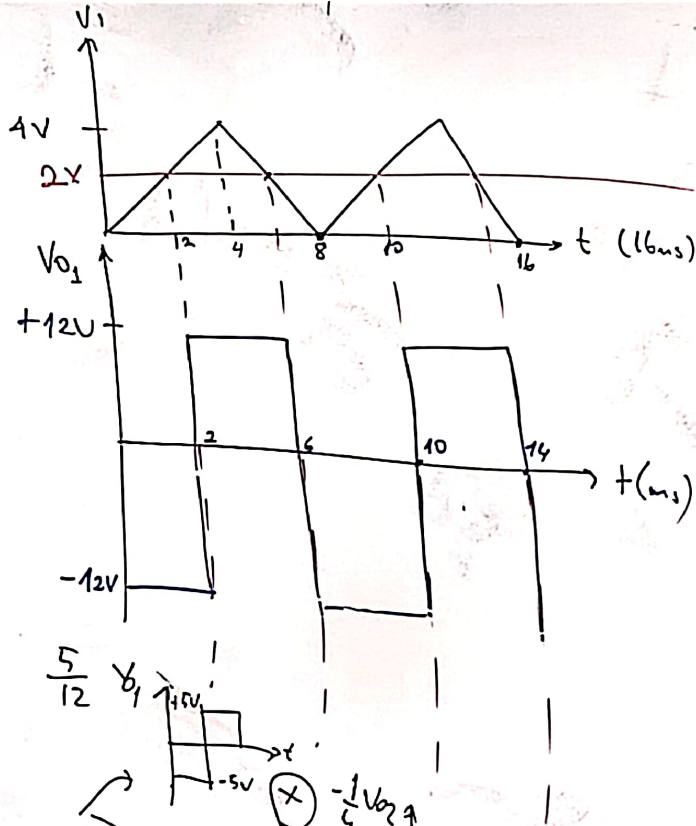
$$T = 16ms$$

V_{O1}, V_{O2}, V_O 'ı özetli olarak çiz.

$$\frac{V_{O2} - V_{O1}/3}{4K} = \frac{V_{O1}/3 - V_O}{1K}$$

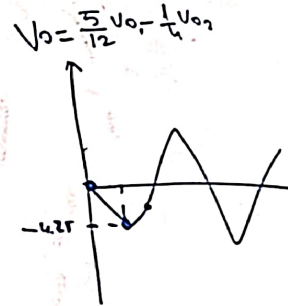
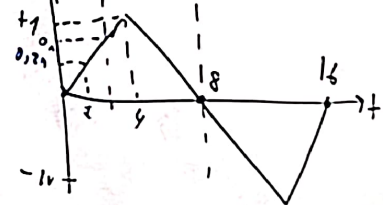
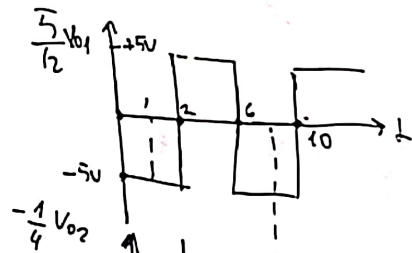
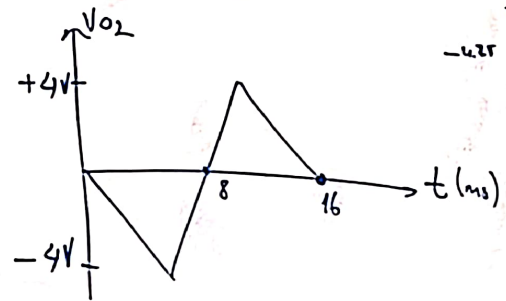
$$V_{O2} - \frac{V_{O1}}{3} = \frac{4}{3}V_{O1} - 4V_O$$

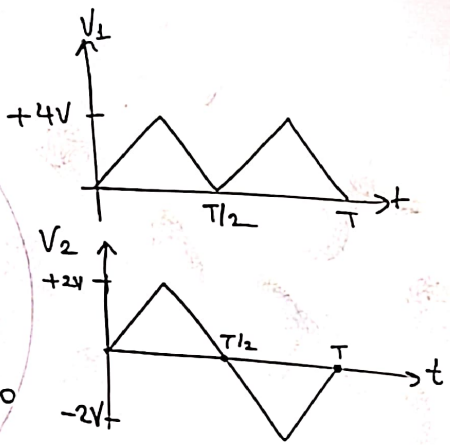
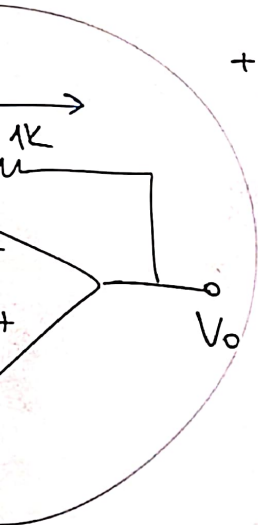
$$4V_O = \frac{5}{3}V_{O1} - V_{O2} \Rightarrow V_O = \frac{1}{4} \left(\frac{5}{3}V_{O1} - V_{O2} \right) = \frac{5}{12}V_{O1} - \frac{1}{4}V_{O2}$$



$$V_{O2} = -V_2 \left(\frac{10K}{5K} \right)$$

$$V_{O2} = -2V_2$$





$\pm V_{cc} = \pm 12V$
 $T = 16ms$

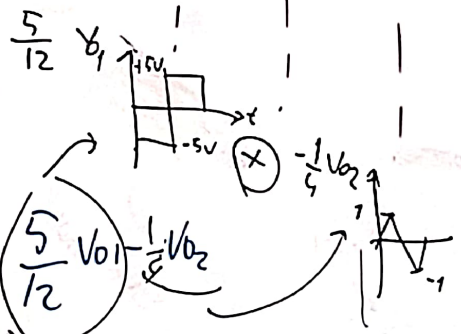
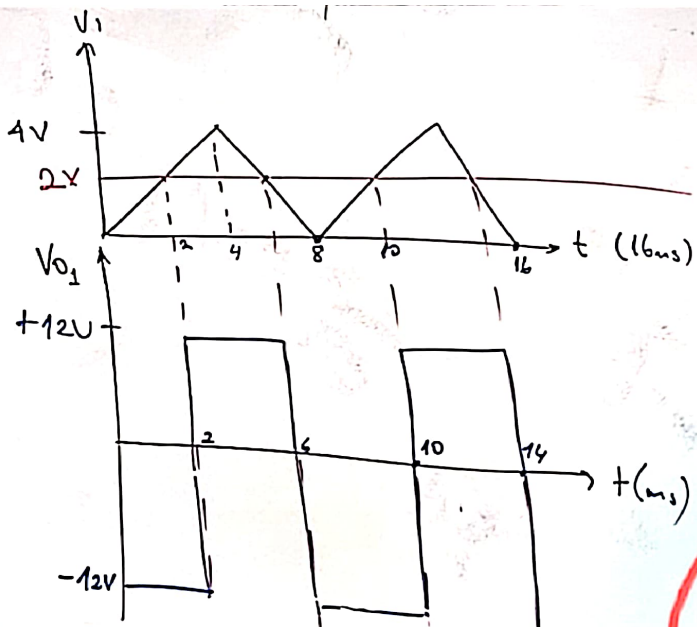
V_{01}, V_{02}, V_0 ötekli olarak çiz.

$$\frac{V_{02} - V_{01}/3}{4K} = \frac{V_{01}/3 - V_0}{1K}$$

$$V_{02} - \frac{V_{01}}{3} = \frac{4}{3}V_{01} - 4V_0$$

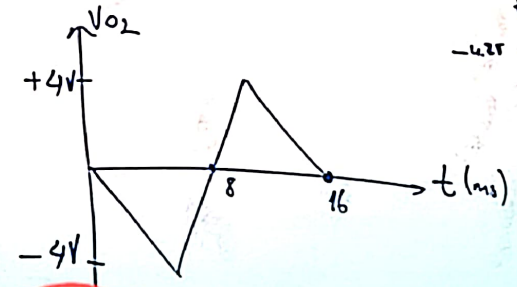
$$4V_0 = \frac{5}{3}V_{01} - V_{02}$$

$$V_0 = \frac{1}{4} \left(\frac{5}{3}V_{01} - V_{02} \right) = \left(\frac{5}{12}V_{01} - \frac{1}{4}V_{02} \right)$$

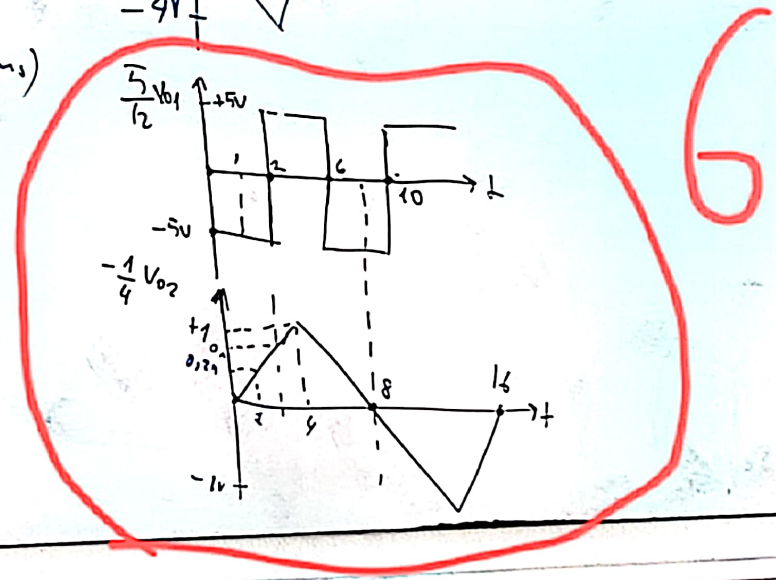
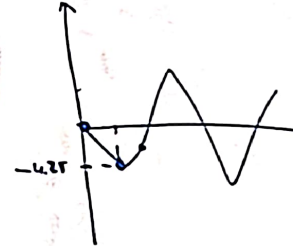


$$V_{02} = -V_2 \left(\frac{10K}{5K} \right)$$

$$V_{02} = -2V_2$$



$$V_0 = \frac{5}{12}V_{01} - \frac{1}{4}V_{02}$$



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